BA-BE Practice 2

Module 7a - Bioavailability and Bioequivalence · Daily practice sheet · No date

Name _____ Date _____

Problem 1 A physician would like an equivalent oral dose of an antibiotic that has been administered to achieve the same extent of absorption as a 500 mg IV bolus dose. The oral drug product has an absolute bioavailability of 67%. What dose would you recommend (Please round to the nearest 10 mg)?

Problem 2 The relative bioavailability of a capsule formulation was studied in 24 volunteers. Each volunteer received either a single oral tablet containing 500 mg of the drug or a capsule containing 500 mg of the drug. The average AUC of the oral tablet was 200 (mcg/mL) hr and the average AUC of the capsule was 160 (mcg/mL) hr. What is the relative bioavailability of the drug from the tablet compared to the capsule?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

Problem 1 Her key

Answer: 750 mg

WHAT THE STEM GIVES

D IV	500 mg	the numerator of D po = $\frac{\text{D IV}}{\text{F}}$
F	67%	F = 0.67, the denominator
rounding	nearest 10 mg	rounds 746.3 to 750 mg

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{F} \times \text{D po} = \text{D IV}, \text{ so D po} = \frac{\text{D IV}}{\text{F}}$$

The same extent of absorption means the oral AUC equals the IV AUC, so the fraction absorbed times the oral dose must equal the whole IV dose.

ALGEBRA

$$\text{D po} = \frac{500 \text{ mg}}{0.67} = 746.3 \text{ mg}$$

F has no units, so the oral dose is in milligrams; it is larger than 500 mg because only 67% of what is swallowed reaches the circulation.

ROUND

$$\text{D po} = 750 \text{ mg}$$

The stem asks for the nearest 10 mg, so 746.3 mg is reported as 750 mg, a dose that can be given. Check: 0.67 × 750 mg = 502.5 mg reaches the circulation, within 2.5 mg of the 500 mg IV dose, so the oral AUC matches the IV AUC to the rounding.

> **DOES THE NUMBER MAKE SENSE?**
> Two thirds of the oral dose is absorbed, so the oral dose is half as much again as 500 mg: about 750 mg, and 746.3 rounds to 750. The oral dose must exceed 500 mg because F is below 1.

Problem 2 Her key

Answer: 1.25 (as a decimal fraction)

WHAT THE STEM GIVES

n	24 volunteers	*not needed*
D tablet	500 mg	the dose ratio $\frac{500}{500} = 1$
D capsule	500 mg	the dose ratio $\frac{500}{500} = 1$
AUC tablet	200 (mcg/mL) hr	the numerator: the tablet is the test product
AUC capsule	160 (mcg/mL) hr	the denominator: the capsule is the reference

THE WORKING, ONE LINE AT A TIME

SETUP

$$F_{rel} = \frac{AUC_{tablet}}{AUC_{capsule}} \times \frac{D_{capsule}}{D_{tablet}}$$

The tablet is compared to the capsule, so the capsule is the reference and its AUC goes in the denominator; the doses are both 500 mg, so the dose ratio is 1.

ALGEBRA

$$F_{rel} = \frac{200\ (mcg/mL)hr}{160\ (mcg/mL)hr} \times \frac{500\ mg}{500\ mg} = 1.25$$

The areas cancel their units and the doses cancel theirs, leaving a pure ratio; 1.25 means the tablet delivers 25% more drug than the capsule.

DOES THE NUMBER MAKE SENSE?

Equal doses, so F rel is the AUC ratio alone: 200 to 160 is 5 to 4, which is 1.25. It is above 1 because the tablet gave the larger area; a value below 1 means the ratio was inverted.